BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Downloads/Cornell/BEE 5750/hw2-jinry`
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

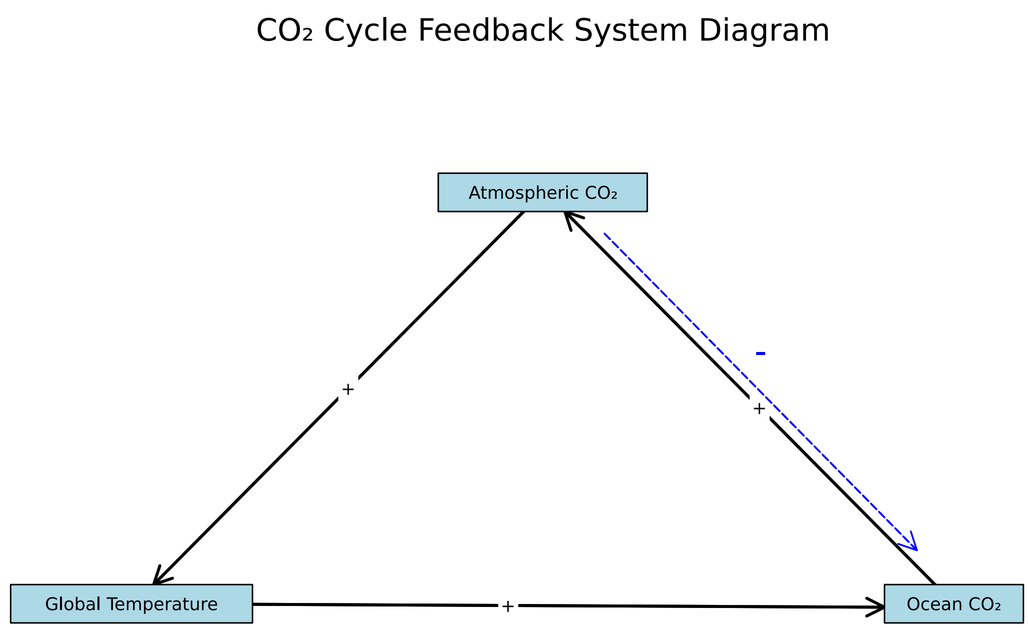
Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```
using Pkg
Pkg.add("GraphRecipes")
```

```
   Resolving package versions...
  No Changes to `~/Downloads/Cornell/BEE 5750/hw2-jinry/Project.toml`
  No Changes to `~/Downloads/Cornell/BEE 5750/hw2-jinry/Manifest.toml`
```

```
using GraphRecipes, Plots

A = [0 1 0;
     0 0 1;
     1 0 0]

names = ["Atmospheric CO ", "Global Temperature", "Ocean CO "]

edge_labels = Dict((1, 2) => "+", (2, 3) => "+", (3, 1) => "+")

shapes = [:rect, :rect, :rect]
xpos = [0, -1, 1]
ypos = [1, 0, 0]

p = graphplot(A, names=names, edgelabel=edge_labels, markersize=0.07,
              markershapes=shapes, markercolor=:lightblue, x=xpos, y=ypos,
              arrow=:true, curves=false, linewidth=2, fontsize=8,
              title="CO  Cycle Feedback System Diagram")
```

```
x_start, y_start = xpos[1] + 0.15, ypos[1] - 0.1
x_end, y_end = xpos[3] - 0.09, ypos[3] + 0.13


plot!([x_start, x_end], [y_start, y_end],
      line=:dash, linewidth=1.2, color=:blue, arrow=true, arrowhead=:triangle, size=(800, 50


annotate!((x_start + x_end)/2, (y_start + y_end)/2 + 0.1,
          text("-", 15, :blue, :center))

display(p)
```



I drew it base on the understanding that when the concentration of atmospheric CO rises, it acts as a greenhouse gas, which leads to an increase in the global mean temperature. This rise in temperature then negatively affects the ocean's ability to store CO , mainly by reducing the solubility of CO in seawater. Consequently, the ocean releases more CO back into the atmosphere, reinforcing this cycle. The dashed arrow is drawn because of Henry's Law. When air has more CO than the ocean, the gas naturally dissolves into the seawater. This helps to lower the CO in the air.

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

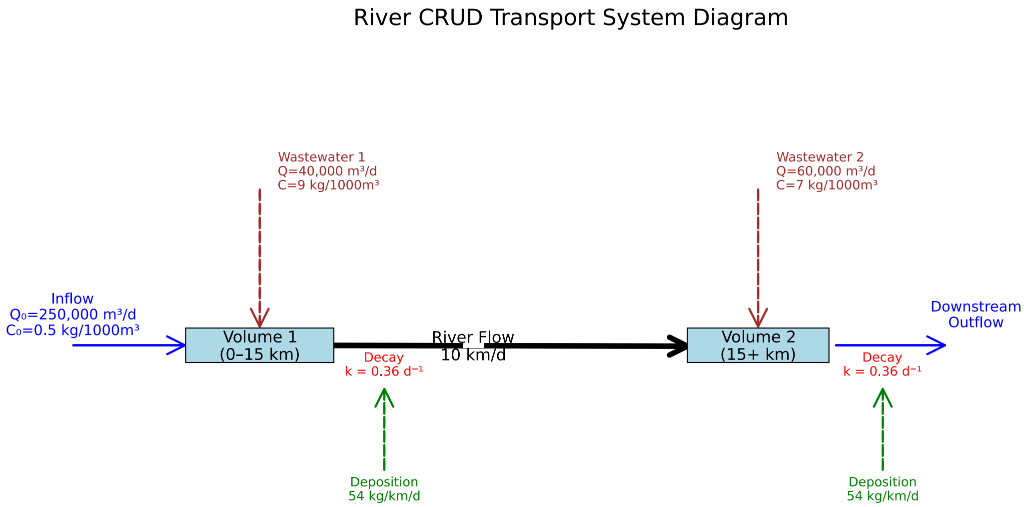
Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

```
using GraphRecipes, Plots

A = [0 1;
     0 0]

names = ["Volume 1\n(0-15 km)",
         "Volume 2\n(15+ km)"]


edge_labels = Dict((1, 2) => "River Flow\n10 km/d")

shapes = [:rect, :rect]
xpos = [0, 8]
ypos = [0, 0]

p = graphplot(A,
              names=names,
              edgelabel=edge_labels,
              markersize=0.3,
              markershapes=shapes,
              markercolor=:lightblue,
              x=xpos, y=ypos,
              arrow=true, curves=false, linewidth=5, fontsize=10,
```

```
              title="River CRUD Transport System Diagram",
              xlim=(-3, 13), ylim=(-3, 5),
              size=(1100, 500))


plot!([-5.25, -1.2], [0, 0], line=:solid, linewidth=2, color=:blue, arrow=true, arrowhead=:ti
annotate!(-3, 0.5, text("Inflow\nQ =250,000 m³/d\nC =0.5 kg/1000m³", 9, :blue, :center))

plot!([9.25, 11], [0, 0], line=:solid, linewidth=2, color=:blue, arrow=true, arrowhead=:trian
annotate!(11.5, 0.5, text("Downstream\nOutflow", 9, :blue, :center))

plot!([0, 0], [2.5, 0.3], line=:dash, linewidth=2, color=:brown, arrow=true, arrowhead=:trian
annotate!(0.3, 2.8, text("Wastewater 1\nQ=40,000 m³/d\nC=9 kg/1000m³", 8, :brown, :left))

plot!([8, 8], [2.5, 0.3], line=:dash, linewidth=2, color=:brown, arrow=true, arrowhead=:trian
annotate!(8.3, 2.8, text("Wastewater 2\nQ=60,000 m³/d\nC=7 kg/1000m³", 8, :brown, :left))

plot!([2, 2], [-2, -0.7], line=:dash, linewidth=2, color=:green, arrow=true, arrowhead=:trian
annotate!(2, -2.3, text("Deposition\n54 kg/km/d", 8, :green, :center))

plot!([10, 10], [-2, -0.7], line=:dash, linewidth=2, color=:green, arrow=true, arrowhead=:tri
annotate!(10, -2.3, text("Deposition\n54 kg/km/d", 8, :green, :center))

annotate!(2, -0.3, text("Decay\nk = 0.36 d ¹", 8, :red, :center))
annotate!(10, -0.3, text("Decay\nk = 0.36 d ¹", 8, :red, :center))

display(p)
```



I decided to include two boxes for control volumes in this system diagram. I divided it into two boxes because the variables to be considered for each box are different. For the first box, it only needs to consider the concentration changes caused by the inflow of water and wastewater 1. For box2, it needs to take into account the water from box1, the concentration changes caused by natural atmospheric deposition, and the concentration changes caused by wastewater2.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproteorozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Left-hand side (C dT/dt): The rate at which the Earth's climate system stores heat. C is the heat capacity, and dT/dt is the rate of temperature change. Right-hand side: The net energy flow into the system. we solve the equation for the derivative dT/dt, which represents the instantaneous rate of temperature change.

To prepare for discretization, I realize i need to solve the equation for the derivative dT/dt, which represents the instantaneous rate of temperature change:

$$\frac{dT}{dt} = \frac{1}{C}\left[\frac{(1-\alpha)S}{4} - (A - BT)\right]$$

Since, we cannot directly equate a rate per year (°C/year) to a rate per second (J/(s · m²)), I multiply the right-hand side by the number of seconds in a year to convert the energy flow from a power (Joules/second) to an annual energy (Joules/year).

$$\frac{dT}{dt} = \frac{1}{C}\left[\frac{(1-\alpha(T))S}{4} - (A - BT)\right] \quad k_{year}$$

We can now apply the Forward Euler Discretization and apply the Forward Euler Discretization. And we already know that delta t is the discrete time step (1 year, as given in the problem). We can find the equation to be

$$T_{n+1} = T_n + \frac{k^{year}}{C}\left[\frac{(1-\alpha(T_n))S}{4} - (A - BT_n)\right]$$

Problem 3.2

Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
using Plots

#
C = 51.0  # J/(m²·°C)
A = 221.2 # W/m²
B = -1.3  # W/(m²·°C)
 _i = 0.5 # ice albedo
 _0 = 0.3 # ocean albedo
S_present = 1368.0 # present solar constant
S_neoproterozoic = 1272.0 # Neoproterozoic solar constant

#
seconds_per_year = 365.25 * 24 * 3600
```

```
#
function albedo(T)
    if T   -10.0
        return  _i
    elseif T   10.0
        return  _0
    else
        return  _i + ( _0 -  _i) * ((T + 10.0) / 20.0)
    end
end

#
function dT_dt(T, S)
    incoming = (1.0 - albedo(T)) * S / 4.0
    outgoing = A - B * T
    net_energy = incoming - outgoing
    return (seconds_per_year / C) * net_energy
end

#
function simulate(T0, S, years, Δt=1.0)
    T_values = Float64[T0]
    T_current = T0

    for t in 1:years
        T_next = T_current + dT_dt(T_current, S) * Δt
        push!(T_values, T_next)
        T_current = T_next
    end

    return T_values
end

# Problem 3.2
simulation_years = 200
initial_temperatures = -60:10:30

#
plt = plot(xlabel="Time (years)", ylabel="Temperature (°C)",
           title="Climate Model Trajectories (Neoproterozoic S=$(S_neoproterozoic))",
           legend=:bottomright,
           size=(800, 600),
```

```
           xlims=(0, simulation_years),
           xticks=0:40:simulation_years,  #   x
           yticks=-60:20:40)              #   y

for T0 in initial_temperatures
    trajectory = simulate(T0, S_neoproterozoic, simulation_years)
    plot!(plt, 0:simulation_years, trajectory, label="T  = $T0", linewidth=2)
end

display(plt)
```

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

References

List any external references consulted, including classmates.